

Charlie Marx

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charliemarx.github.io

EDUCATION

Stanford University
Ph.D. in Computer Science
Advisor: Stefano Ermon

2020 - PRESENT

Haverford College
B.S. in Computer Science, *highest honors*
B.S. in Mathematics, *highest honors*
GPA: 3.96/4.0, *magna cum laude*

2016 - 2020

AWARDS & HONORS

NSF Graduate Research Fellowship
Stanford School of Engineering Fellowship
Phi Beta Kappa
CRA Outstanding Undergraduate Researcher Runner-Up
Arnold and Mabel Beckman Foundation Award
Farmers Insurance Standout Student of the Year

2021
2020
2020
2019
2018
2016

SELECTED WORK

[Calibrated Probabilistic Forecasting for Arbitrary Sequences](#)
Charles Marx, Volodymyr Kuleshov, Stefano Ermon
arXiv Preprint, 2024

[Calibration by Distribution Matching: Trainable Kernel Calibration Metrics](#)
Charles Marx*, Sofian Zalouk*, Stefano Ermon
NeurIPS – *Conference on Neural Information Processing Systems*, 2023

[Predictive Multiplicity in Classification](#)
Charles Marx, Flavio du Pin Calmon, Berk Ustun
ICML – *International Conference on Machine Learning*, 2020
Previously Selected for Oral Presentation at 2019 NeurIPS HCML Workshop

PUBLICATIONS

1. [Online Calibrated and Conformal Prediction Improves Bayesian Optimization](#)
Shachi Deshpande, **Charles Marx**, Volodymyr Kuleshov
AISTATS – *Artificial Intelligence and Statistics*, 2024
2. [Calibration by Distribution Matching: Trainable Kernel Calibration Metrics](#)
Charles Marx*, Sofian Zalouk*, Stefano Ermon
NeurIPS – *Conference on Neural Information Processing Systems*, 2023
3. [But Are You Sure? An Uncertainty-Aware Perspective on Explainable AI](#)
Charles Marx, Youngsuk Park, Hilaf Hasson, Yuyang Wang, Stefano Ermon, Jun Huan
AISTATS – *Artificial Intelligence and Statistics*, 2023
4. [Classification with Conceptual Safeguards](#)
Hailey Joren, **Charles Marx**, Berk Ustun
ICLR – *International Conference on Learning Representations*, 2023
5. [Modular Conformal Calibration](#)
Charles Marx*, Shengjia Zhao*, Willie Neiswanger, Stefano Ermon
ICML – *International Conference on Machine Learning*, 2022
6. [Predictive Multiplicity in Classification](#)
Charles Marx, Flavio du Pin Calmon, Berk Ustun
ICML – *International Conference on Machine Learning*, 2020
Previously Selected for Oral Presentation at 2019 NeurIPS HCML Workshop
7. [Disentangling Influence: Using Disentangled Representations to Audit Model Predictions](#)
Charles Marx, Richard Phillips, Sorelle Friedler, Carlos Scheidegger, Suresh Venkatasubramanian

	NeurIPS – <i>Conference on Neural Information Processing Systems</i> , 2019	
8.	Examining Inputs of Biogenic and Oil-Derived Hydrocarbons in Surface Waters Following the Deepwater Horizon Oil Spill Helen White, Charles Marx , David Valentine, Charles Sharpless, Christoph Aeppli, Kelsey Gosselin, Veronika Kivenson, Rachel Liu, Robert Nelson, Sean Sylva, Christopher Reddy <i>ACS Earth and Space Chemistry</i> , 2019	
9.	Rapid Identification of Deepwater Horizon Oil Residues Using X-Ray Fluorescence Anna Michel, Alexandra E. Morrison, Charles Marx , Helen White <i>ACS Environmental Science & Technology Letters</i> , 2018	
RESEARCH & INDUSTRY EXPERIENCE	Amazon (Language Modeling) <i>Applied Science Intern</i> Designed algorithms for detecting hallucinations in large language models. <i>Advisors:</i> Wooseok Ha and Christian Bock	SUMMER 2023
	Amazon (Time Series ML) <i>Applied Science Intern</i> Designed machine learning methods for quantifying uncertainty in model explanations. <i>Advisors:</i> Youngsuk Park and Hilaf Hasson	SUMMER 2022
	International Food Policy Research Institute <i>Intern</i> Built computer vision systems to classify land usage in Africa using satellite images. <i>Advisors:</i> Channing Arndt and Zhe Guo	SUMMER 2020
	Harvard University <i>Research Assistant</i> Developed integer programming tools to measure underspecification in machine learning tasks. <i>Advisors:</i> Flavio Calmon and Berk Ustun	SUMMER 2019
TEACHING	Stanford CS 228: Probabilistic Graphical Models (Head TA)	2024
	Stanford CS 229: Machine Learning	2024
	Haverford CMSC 360: Machine Learning	2020
INVITED & CONTRIBUTED TALKS	Data for Sustainability Conference <i>Stanford, CA</i> Probabilistic Forecasting for Science <i>Best Talk Award</i>	2023
	ICML Conference <i>Baltimore, MD</i> Modular Conformal Calibration	2022
	Simons Institute for the Theory of Computing <i>Berkeley, CA</i> Disentangling Influence	2019
	INFORMS Annual Meeting <i>Seattle, WA</i> Predictive Multiplicity in Classification	2019
	FAT* Conference <i>New York, NY</i> Auditing Black Box Models (60-minute tutorial) <i>with Carlos Scheidegger and Suresh Venkatasubramanian</i>	2018
	GOMOSSES Conference <i>New Orleans, LA</i> Examining Inputs of Biogenic and Oil-Derived Hydrocarbons	2018
ACADEMIC SERVICE	Reviewing: NeurIPS ('22, '24), ICML ('22), KDD ('20), IEEE JSAT ('23), ACM JRC ('23) Program Committee: FAccT (2022, 2024), NeurIPS Workshop on Fair AI in Finance (2020)	
EXTRA- CURRICULAR ACTIVITIES	Captain of Haverford College Men's Club Soccer	2018 – 2019
	Leader of Haverford College Outdoors Club	2017 – 2018
	President of Furniture for Newly Housed Families Organization	2015 – 2017
SKILLS & INTERESTS	Languages: <i>Fluent:</i> Python, R, Bash · <i>Familiar:</i> SQL, C Tools: PyTorch, Tensorflow, Pandas, CPLEX Interests: Backpacking, Basketball, Cooking, Soccer	